

NYC School Bus Delays and Breakdowns

What causes school bus delays across New York City, which operators and boroughs are worst affected, and how the pattern moves through the week.

Tools Excel. Pivot tables, charts, data cleaning

Data NYC Bus Breakdown and Delays. 282,191 reported incidents

The questions

The New York Division of Transportation wanted to understand why the school bus network was generating so many delays and breakdowns, and where to focus to reduce them:

- What are the most common reasons for delays and breakdowns?
- How do delay times vary by bus company and borough?
- Is there a correlation between specific days of the week and the frequency of breakdowns or delays?

The data

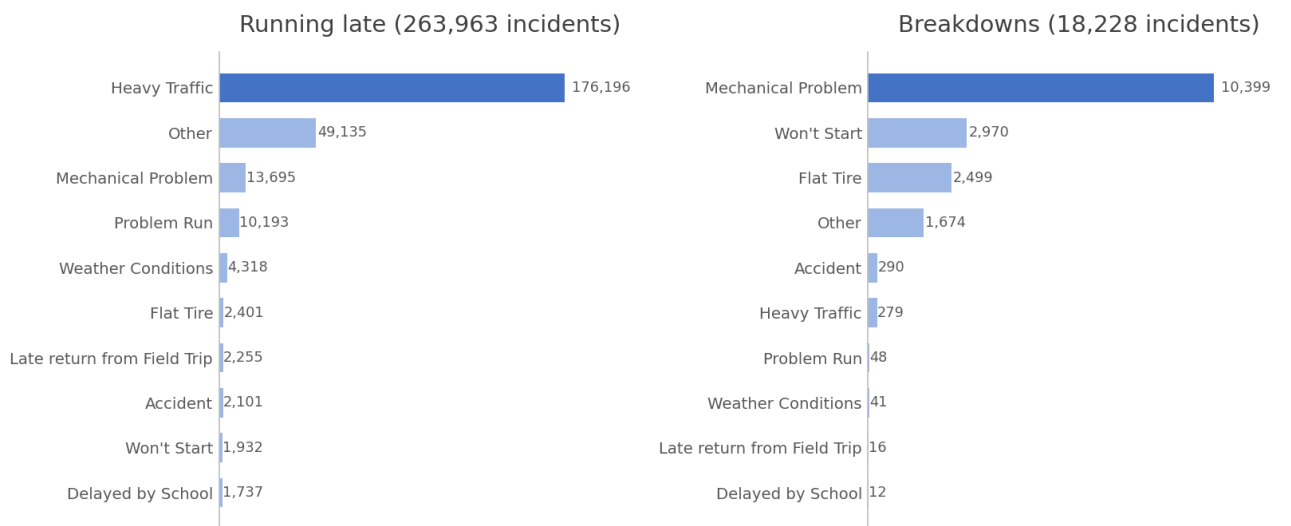
Each row is one reported incident, logged by the operator with a reason, a borough, a bus company, a date and a free-text estimate of how long the delay lasted. Three things had to be fixed before any of it could be measured.

Problem	What I did
Delay length was free text, entered inconsistently as ranges, single numbers and phrases	Split it into a lower and upper bound in minutes, so delays could be averaged instead of counted
Bus company names appeared under several spellings and formats	Standardised them into a single clean company field so each operator aggregates as one
No weekday field existed	Derived day of week from the incident date to test the weekly pattern

Incidents split into two types that behave very differently and are kept separate throughout: 263,963 running late, and 18,228 actual breakdowns.

Why buses are late, and why they break down

These are two different problems. Delays are overwhelmingly a traffic story: heavy traffic alone accounts for 176,196 incidents, roughly two thirds of everything reported. Breakdowns are a fleet story: mechanical problems, buses that will not start and flat tires make up 87% of them, and traffic barely appears.



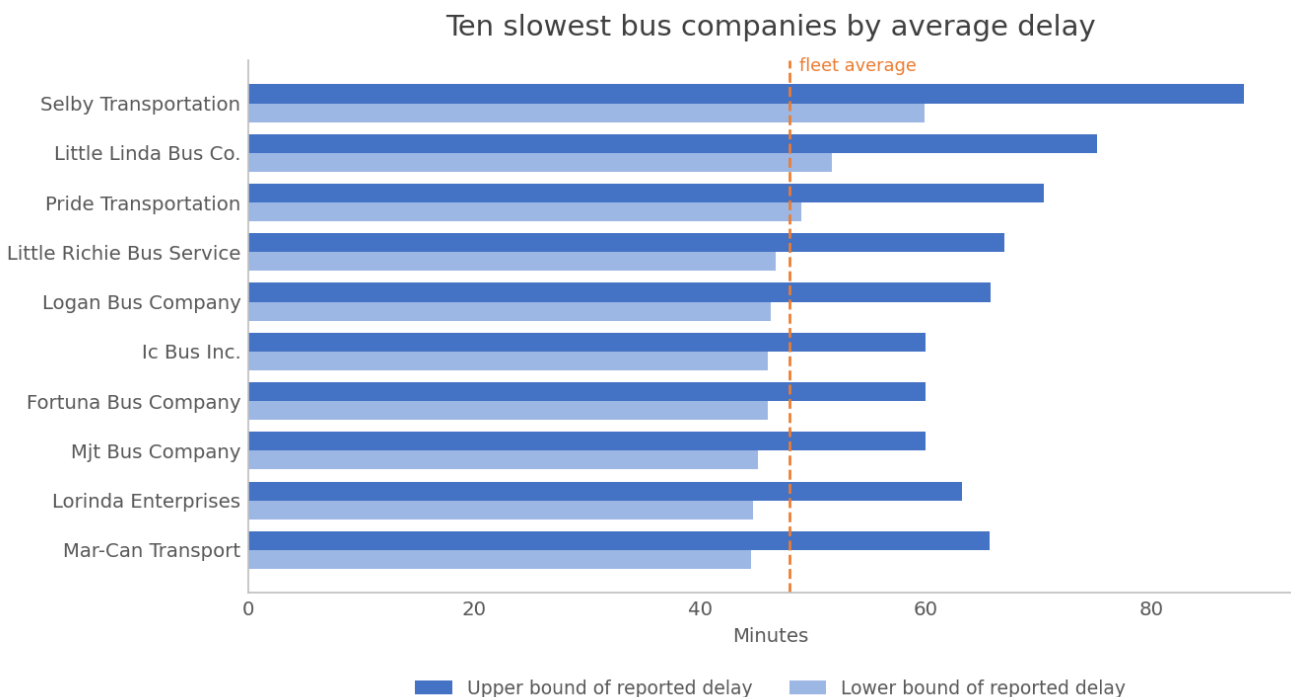
Reported reasons by incident type. Note the two panels use different scales.

The practical read is that delays and breakdowns need separate remedies. Nothing done to the fleet will move the delay number much, and nothing done about traffic will prevent breakdowns.

One caveat worth flagging: "Other" is the second largest delay category at 49,135 incidents and the fourth largest breakdown category. That is a reporting problem rather than a finding, and tightening the reason list would make future analysis considerably sharper.

Delay times by operator

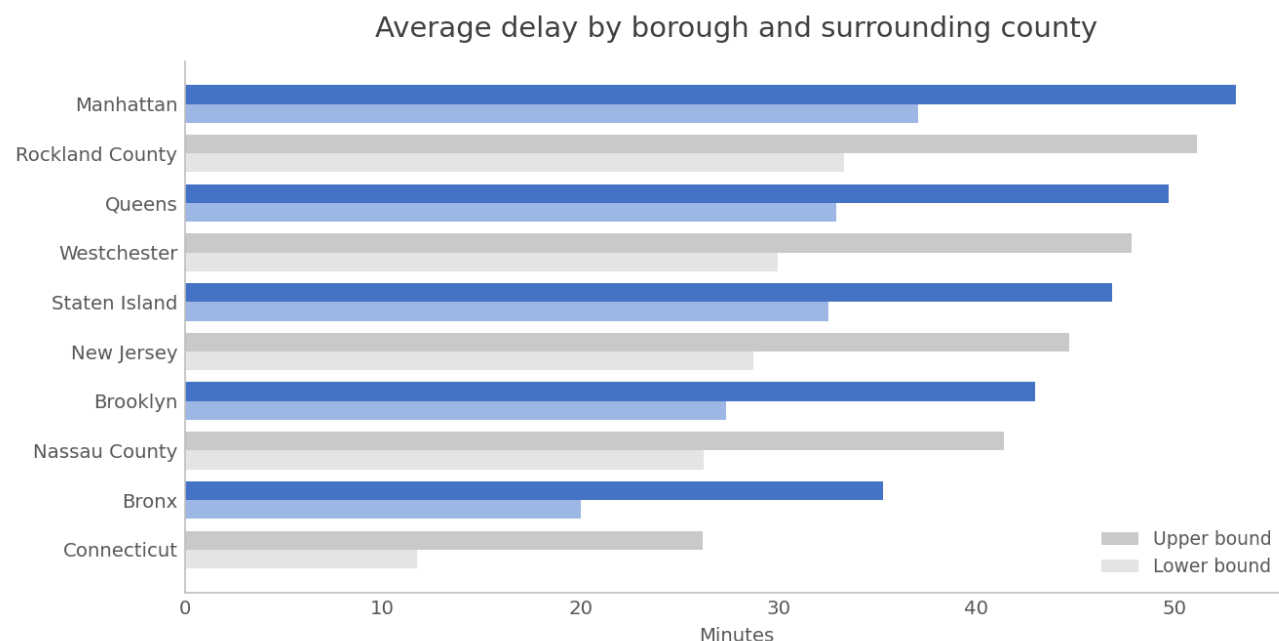
Averaged across the fleet, a reported delay runs between 48 and 69 minutes. The ten slowest operators sit above that, and Selby Transportation is a clear outlier at 60 to 88 minutes, roughly 28% worse than the fleet average at the upper bound.



Ten operators with the highest average delay, against the fleet average.

Delay times by borough

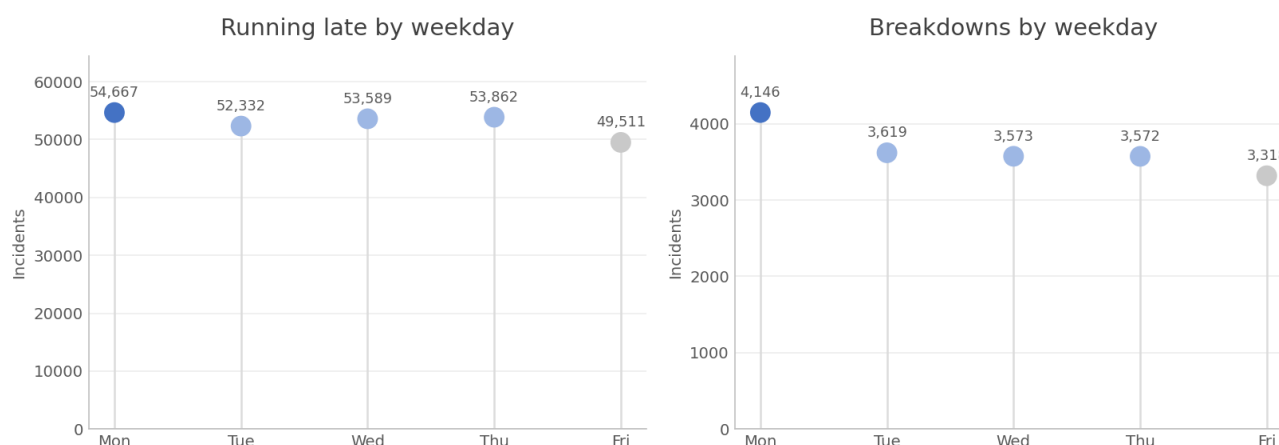
Delays are longest in the densest parts of the city. Manhattan averages 37 to 53 minutes, against 20 to 35 in the Bronx and 12 to 26 in Connecticut. The ordering tracks population and vehicle density closely, which is consistent with traffic being the dominant cause.



Average delay by location. Blue marks the five NYC boroughs.

The weekly pattern

Both measures peak on Monday and fall to their lowest on Friday. Delays drop 9% from Monday to Friday and breakdowns drop 20%. Since traffic drives most delays, the Friday dip is consistent with fewer people commuting at the end of the week.



Incidents by weekday. Both panels start at zero, so the differences are real but modest.

Interpretation and limitations

If the goal is fewer delays, the levers are routing and scheduling in Manhattan, Queens and Staten Island, and building extra time into Monday runs. If the goal is fewer breakdowns, the lever is fleet maintenance, and it is worth asking whether the operators at the top of the delay table are also the ones running the oldest

vehicles.

Two limitations matter. Delay length is self-reported by operators and was recorded as free text, so the minute figures are estimates rather than measurements, and an operator that consistently rounds up will look worse than one that rounds down. And this counts incidents, not rates. A company running far more routes will report more delays without necessarily performing worse, so the company comparison uses average delay length rather than incident counts for that reason.

What is in the workbook

Sheet	Contents
Scenario	The brief and the three questions
Bus_Breakdown_and_Delays_NYC	The original file, untouched
Cleaned Data	Adds split delay bounds, clean company name and weekday
Common Delay Reasons	Reason pivots and charts for both incident types
Delay Times	Company and borough pivots and charts
Day Week	Weekday pivots and charts

Portfolio project built on public NYC school bus incident data. The Division of Transportation client in the brief is a scenario used to frame the analysis, not a real engagement.